

# Matcap Maker Usage Guide

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Matcap Maker is a tool for creating Matcap (Material Capture) textures for 3D models. By layering multiple effects (lights, fresnel, noise, images, etc.), you can intuitively create original Matcaps.

## Language Support

This application supports both English and Japanese. You can switch languages from the "Options" menu. Please note that changing the language requires restarting the application.

## Interface Overview

The application is divided into three main areas:

### 1. Preview Window (Center)

- Displays a real-time rendering of your Matcap on a sphere.
- Use the "Export Image" button at the bottom to save the current result as an image file (PNG/JPG).

### 2. Layer List (Left)

- Lists all layers currently in the project.
- Processing occurs from top to bottom.
- **Add:** Create a new layer using the "Add Layer" button.
- **Remove:** Delete the currently selected layer.
- **Toggle Visibility:** Click the circular button next to each item to enable/disable the layer.
- **Reorder:** Drag and drop layers to change their order.
- **Context Menu:** Right-click on a layer to access "Duplicate" or "Delete" options.

### 3. Properties Panel (Right)

- Adjust detailed settings for the layer selected in the list.

## Layer Types and Properties

### 1. Base Layer

Sets the foundation color and preview settings for the Matcap.

- **Color:** Sets the base color.
- **Preview Options:**
  - **Mode:**
    - **Standard:** Standard sphere preview.
    - **With Normal Map:** Previews how the Matcap looks with a normal map applied.
  - **Normal Map:** Select a normal map image to load.
  - **Strength/Scale/Offset:** Adjust the application of the normal map.

### 2. Spot Light

Adds lighting effects from a specific direction. Ideal for creating highlights.

- **Blend Mode:** Select how the light blends (Add, Screen, Overlay, etc.).

- **Intensity:** The brightness of the light.
- **Range:** The spread of the light.
- **Blur:** The softness of the light edges.
- **Scale X/Y:** Changes the aspect ratio of the light shape.
- **Rotation:** Rotates the light shape.
- **Direction X/Y/Z:** Sets the direction the light is coming from.
- **Color:** The color of the light.

### 3. Fresnel / Rim

Adds light to the contours (edges) of the object. Used for rim lights and material definition.

- **Intensity:** The brightness of the light.
- **Power:** The sharpness of the rim. Higher values push the light closer to the edge.
- **Bias:** Offsets the range of the effect.
- **Color:** The color of the light.

### 4. Noise

Adds overall grain or texture.

- **Intensity:** The strength of the noise.
- **Scale:** The fineness of the noise grain.
- **Seed Offset:** Changes the random seed for the noise pattern.
- **Color:** The color of the noise. Usually used with "Multiply" blend mode for shading.

### 5. Image

Maps an arbitrary image as a texture. Used for reflections or patterns.

- **Image:** Select the image file to use.
- **Mapping:**
  - **UV:** Maps the image according to the sphere's UV coordinates.
  - **Planar:** Maps the image like a planar projection.
- **Offset X/Y:** Adjusts the position of the image.
- **Scale:** Adjusts the size of the image.
- **Rotation:** Adjusts the angle of the image.
- **Blur:** Blurs the image.
- **Opacity:** Adjusts the transparency of the image.

### 6. Color Adjustment

Adjusts the overall tone of the composite result up to this layer.

- **Hue:** Adjusts the hue.
- **Saturation:** Adjusts the color intensity.
- **Brightness:** Adjusts the brightness.
- **Contrast:** Adjusts the contrast.

## Menu Operations

## File

- **Open Project:** Opens a saved project file (.json).
- **Save Project:** Saves the current state as a project. Used image assets are also copied to an **assets** folder.
- **Export Image:** Exports the Matcap image, same as the button at the bottom.

## Options

- **Language:** Switches language between English and Japanese (Release requires restart).
- **Resolution:** Selects the export resolution (64x64 to 4096x4096).
- **Padding:** Sets the margin for the exported image.

## Help

- **Third Party Notices:** Displays legal notices for libraries used.